

Практикум E8:archive

Thirty-one questions, 100 marks. Every question the archive asks about the shape of a graph, from May 2021 to November 2025, in the order of the eight techniques rather than the order of the papers.

О практикуме

Тема взята целиком: 43 блока и 114 баллов, внутри границ гранулярности. 39 блоков и 101 балл рабочих — ноябрь 2023 года задвоился и здесь: Paper 1 TZ1 и TZ2 оказались одной бумагой (вопрос 11 совпадает слово в слово), а Paper 3 той сессии общий. Четыре блока прогнаны классификацией дважды, 13 баллов.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e8-stationary-points

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

E8 archive — stationary points, all of them

Thirty-one questions, 100 marks.

Every question the archive asks about the shape of a graph, from May 2021 to November 2025, in the order of the eight techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-e8-stationary-points.ipynb`.

§	technique	questions	marks
1	Find the point	8	27
2	Name the kind from f''	4	13
3	Name the kind from the sign of f'	2	7
4	The point of inflexion	8	18
5	A family with a letter	2	9
6	Which side of the axis	2	2
7	How many	3	16
8	There are none	2	8

The checks are the same ones the practicum uses and they store nothing. Each is handed the curve exactly as the question names it and then **walks along it**: `verify_turning` finds every flat place and compares, `verify_nature` looks at the neighbours, `verify_bend` measures the bend with a chord, `verify_side` reads the sign of the second coordinate, `verify_param_set` and `verify_condition` ask the property itself at a grid of values. Three significant figures, unless the question asks otherwise.

Parts of one question are split by technique. The Paper 3 cubic of May 2021 TZ1 appears in §§ 5, 6 and 7; the Paper 3 family of November 2023 in §§ 2, 5 and 6; the implicit curve of May 2022 TZ2 in §§ 4 and 8. Each part repeats what it needs.

A *show that* is checked by substance, not by the printed line: a point claimed to be stationary has to be flat on the curve itself, a point claimed to be a maximum has to have the curve lower on

both sides, a point claimed to be an inflexion has to have the chord change sides.

Answers that are words go in quotes:

```
'maximum', 'minimum', 'inflexion', 'above',  
'below'. Sets go as FiniteSet(0),  
Interval.open(0, oo).
```

Solutions are at the very bottom, deliberately far away.

```
import sys  
sys.path.append('..')          # from  
practicum/calculus to practicum/kit/  
import sympy as sp             # the escape  
hatch: anything not in kit is in sp  
from kit import *              # checks +  
curve, stationary, nature, crossings  
  
language('en')                 # this  
notebook is in English, and so are the checks  
  
print('ready; sympy', sp.__version__)
```

§ 1. Find the point

Eight questions, 27 marks. Solve $f'(x) = 0$, then go back to f for the second coordinate. Both coordinates carry marks, and so does finding *every* such point.

1.1 — May 2025 TZ1 Paper 1 Q1, 5 marks

Consider $f(x) = \frac{4x^3}{3} - 16x$ for $x \in \mathbb{R}$. The graph of $y = f(x)$ has a local minimum point at (p, q) where $p > 0$.

Find the value of p and the value of q .

```
q1_1 = ...          # the point (p, q)

verify_turning('1.1', q1_1, 4*x**3/3 - 16*x,
               'minimum', domain=(0, 6))
```

1.2 — November 2023 TZ2 Paper 1 Q11(a), 5 marks

Consider $f(x) = e^{\cos 2x}$, where $-\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$.

Find the coordinates of the points on the curve $y = f(x)$ where the gradient is zero.

```
q1_2 = [...]      # every point of zero
gradient

verify_turning('1.2', q1_2, exp(cos(2*x)),
domain=(-pi/4, 5*pi/4))
```

1.3 — November 2022 Paper 2 Q2(a), 2 marks

The function f is defined as $f(x) = \ln(xe^x + 1) - x^4$ for $0 \leq x \leq 2$. The graph of f has a local maximum at point A .

Find the coordinates of A .

```
q1_3 = ...        # the coordinates of A

verify_turning('1.3', q1_3, log(x*exp(x) + 1)
- x**4, 'maximum', domain=(0, 2))
```

1.4 — May 2025 TZ2 Paper 2 Q12(c), 3 marks

The curve C has equation $4x^2 + y^2 - 24x + 4y + 20 = 0$, and its domain is $3 - \sqrt{5} \leq x \leq 3 + \sqrt{5}$. The curve has a maximum point at A .

Find (x_A, y_A) , the coordinates of A .

```
q1_4 = ...          # the coordinates of A

shape = curve(Eq(4*x**2 + y**2 - 24*x + 4*y +
20, 0))

verify_turning('1.4', q1_4, shape, 'maximum',
domain=(3 - sqrt(5), 3 + sqrt(5)))
```

1.5 — May 2025 TZ2 Paper 3 Q1(a) (iii), 2 marks

Consider the curve given by $y = \frac{x(x^2 - 16)}{x^2 + 16}$.

State the coordinates of the local maximum point and the coordinates of the local minimum point.

```
q1_5 = [...]        # the local maximum and the
local minimum

verify_turning('1.5', q1_5, x*(x**2 -
16)/(x**2 + 16), domain=(-10, 10))
```

1.6 — May 2023 TZ1 Paper 3 Q2(b), 2 marks

Consider two numbers $x_1, x_2 \in \mathbb{R}^+$ with $x_1 + x_2 = 12$, so that their product is $f(x_1) = x_1(12 - x_1)$.

(i) Find the value of x_1 for which the function is maximum.

(ii) Hence show that the maximum product of x_1 and x_2 is 36.

Enter both as one point: the value of x_1 and the product there.

```
q1_6 = ...          # (the best x1, the product
there)

verify_turning('1.6', q1_6, x*(12 - x),
'maximum', domain=(0, 12))
```

1.7 — November 2023 TZ1 Paper 2 Q4(a), 2 marks

A particle moves along a straight line. Its displacement, s metres, from a fixed point O after time t seconds is given by

$$s(t) = 4.3 \sin(\sqrt{3t+5}), \quad 0 \leq t \leq 10$$

The particle first comes to rest after q seconds. Find the value of q .

```
q1_7 = ...          # the value of q

verify_turning('1.7', q1_7, 4.3*sin(sqrt(3*t
+ 5)), var=t, domain=(0, 10),
coordinates=False)
```

1.8 — May 2022 TZ2 Paper 1 Q6(b), 6 marks

A function f is defined by $f(x) = x\sqrt{1-x^2}$ where $-1 \leq x \leq 1$. The range of f is $a \leq y \leq b$, where $a, b \in \mathbb{R}$.

Find the value of a and the value of b .

Enter the range as an interval.

```
q1_8 = ...          # the range of f

verify_range('1.8', q1_8, x*sqrt(1 - x**2),
domain=Interval(-1, 1))
```

§ 2. Name the kind from f''

Four questions, 13 marks. Substitute the stationary point into the second derivative and read the sign. Markschemes for this technique want the value, not only the sign.

2.1 — November 2023 TZ2 Paper 1 Q11(b), 4 marks

For $f(x) = e^{\cos 2x}$ on $-\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$, using the second derivative at each point found in part (a), show that the curve $y = f(x)$ has two local maximum points and one local minimum point.

Give the three kinds as a list, in the order $x = 0, \frac{\pi}{2}, \pi$.

```
q2_1 = [...]      # the kind of each point

verify_nature('2.1', q2_1, exp(cos(2*x)), [0,
pi/2, pi], domain=(-pi/4, 5*pi/4))
```

2.2 — November 2023 Paper 3 Q1(f) (i), 3 marks

Consider the family of curves $y = x^3 + ax^2 + b$ with $a \neq 0$. Its points of zero gradient are $P(0, b)$ and $Q\left(-\frac{2a}{3}, \frac{4a^3}{27} + b\right)$. Consider them for $a > 0$ and $b > 0$.

Find an expression for $\frac{d^2y}{dx^2}$ and hence determine whether each point is a local maximum or a local minimum.

Two answers: the second derivative, and the two kinds with P first.

```
a, b = symbols('a b')

q2_2a = ...      # d^2y/dx^2
q2_2b = [...]    # the kind of each point, in
order

family = x**3 + a*x**2 + b
positive = {a: (3, 1, 2, 6), b: (1, 2, 5, 1)}
# a > 0, b > 0

verify_derivative('2.2a', q2_2a, family,
order=2)
verify_nature('2.2b', q2_2b, family, [0,
-2*a/3], domain=(-8, 4), params=positive)
```

2.3 — November 2023 Paper 3 Q1(g)(i), 1 mark

For the same family, consider the points P and Q for $a < 0$ and $b > 0$.

State whether P is a local maximum or a local minimum.

```
q2_3 = ...          # the kind of the point at
the origin when a < 0

negative = {a: (-3, -1, -2, -6), b: (1, 2, 5,
1)} # a < 0, b > 0

verify_nature('2.3', q2_3, x**3 + a*x**2 + b,
0, domain=(-4, 8), params=negative)
```

2.4 — May 2025 TZ3 Paper 3 Q2(d), 5 marks

A box is folded from an $a \times b$ sheet with squares of side x cut from the corners, so that $V = x(a - 2x)(b - 2x)$ for $0 \leq x \leq \frac{a}{2}$, where $a < b$. The only solutions of $\frac{dV}{dx} = 0$ are $x = \frac{(a + b) \pm \sqrt{a^2 - ab + b^2}}{6}$, and

$$x_m = \frac{(a + b) - \sqrt{a^2 - ab + b^2}}{6}$$

is the one inside the interval.

Use $\frac{d^2V}{dx^2}$ to show that there is a local maximum of V at $x = x_m$.

Two answers: the second derivative, and the kind of point.

```
a, b = symbols('a b', positive=True)

q2_4a = ...      # d2V/dx2
q2_4b = ...      # the kind of point at x_m

V = x*(a - 2*x)*(b - 2*x)
x_m = ((a + b) - sqrt(a**2 - a*b + b**2))/6
sheets = {a: (3, 2, 4, 5), b: (5, 7, 9, 6)}
# a < b, both positive

verify_derivative('2.4a', q2_4a, V, order=2)
verify_nature('2.4b', q2_4b, V, x_m, domain=
(0, a/2), params=sheets)
```

§ 3. Name the kind from the sign of f'

Two questions, 7 marks. The gradient changes from $+$ to $-$, or it does not change at all. This is the technique that still works when f'' is zero.

3.1 — November 2025 TZ1 Paper 1 Q9(b), 2 marks

The function f has a derivative given by $f'(x) = 3x^2 + 12x - 15$, and the graph of $y = f(x)$ has horizontal tangents where $x = a = -5$ and $x = b = 1$. The sign of f' on the three intervals they cut is $+$, then $-$, then $+$.

State, with a reason, whether there is a local maximum point or a local minimum point on the graph of $y = f(x)$ at $x = a$.

```
q3_1 = ...          # the kind of point at x = a

shape = integrate(3*x**2 + 12*x - 15, x)
# any antiderivative will do

verify_nature('3.1', q3_1, shape, -5, domain=
(-9, 5))
```

3.2 — May 2021 TZ2 Paper 3 Q1(g), 5 marks

Consider $f_n(x) = x^n(a - x)^n$ for $a \in \mathbb{Z}^+$ and $n \in \mathbb{Z}^+$, $n > 1$. Earlier parts give $f'_n(x) = nx^{n-1}(a - 2x)(a - x)^{n-1}$ and the fact that $f'_n\left(\frac{a}{4}\right) > 0$.

By using that fact and considering the sign of $f'_n(-1)$, show that the point $(0, 0)$ on the graph of $y = f_n(x)$ is

- (i)** a local minimum point for even values of n , where $n > 1$;
- (ii)** a point of inflexion with zero gradient for odd values of n , where $n > 1$.

```

n, a = symbols('n a', positive=True)

q3_2a = ...      # the kind of point at the
origin for even n
q3_2b = ...      # the kind of point at the
origin for odd n

f_n = x**n*(a - x)**n

verify_nature('3.2(i)', q3_2a, f_n, 0,
domain=(-1, a + 1),
           params={n: (2, 4, 2, 4), a: (2,
2, 3, 5)})
verify_nature('3.2(ii)', q3_2b, f_n, 0,
domain=(-1, a + 1),
           params={n: (3, 5, 3, 5), a: (2,
2, 3, 5)})

```

§ 4. The point of inflexion

Eight questions, 18 marks. $f'' = 0$ and the concavity changes. On a calculator paper the shortest route is a stationary point of f' .

4.1 — May 2021 TZ1 Paper 2 Q11(c), 2 marks

The function f is defined by $f(x) = \frac{3x + 2}{4x^2 - 1}$ for $x \neq \pm\frac{1}{2}$. The graph of $y = f(x)$ has exactly one point of inflexion.

Find the x -coordinate of the point of inflexion.

```
q4_1 = ...      # the x-coordinate of the
point of inflexion

verify_bend('4.1', q4_1, (3*x + 2)/(4*x**2 -
1), domain=(-4, -0.55))
```

4.2 — May 2022 TZ1 Paper 3 Q2(g)(i), 2 marks

Consider the curve $y = (x - r)(x^2 - 2ax + a^2 + b^2)$ for $a \neq r$ and $b > 0$. It has a point of inflexion at P .

Show that the x -coordinate of P is $\frac{1}{3}(2a + r)$. You are not required to demonstrate a change in concavity.

```
a, r, b = symbols('a r b')

q4_2 = ...          # the x-coordinate of the
inflexion, in terms of a and r

cubic = (x - r)*(x**2 - 2*a*x + a**2 + b**2)
shapes = {a: (4, 2, 5, 1), r: (1, -2, 0, 3),
b: (1, 4, 2, 2)}

verify_bend('4.2', q4_2, cubic, domain=(-8,
8), params=shapes)
```

4.3 — May 2022 TZ1 Paper 3 Q2(g) (ii), 1 mark

The points $A(a, g(a))$ and $R(r, 0)$ lie on the same curve.

Hence describe numerically the horizontal position of the point P relative to the horizontal positions of the points R and A .

Enter the x -coordinate of P written as $r + k(a - r)$, that is as a fraction k of the way from R to A .

```
q4_3 = ...          # the same x-coordinate
written as  $r + k(a - r)$ 

verify_bend('4.3', q4_3, cubic, domain=(-8,
8), params=shapes)
```

4.4 — May 2022 TZ1 Paper 3 Q2(h)(ii), 1 mark

Consider the special case where $a = r$ and $b > 0$.

State, in terms of r , the coordinates of the point P .

```
q4_4 = ...          # the coordinates of the
inflexion when  $a = r$ 

same = (x - r)*(x**2 - 2*r*x + r**2 + b**2)

verify_bend('4.4', q4_4, same, domain=(-8,
8), coordinates=True,
              params={r: (1, -2, 3), b: (2, 1,
4)}))
```

4.5 — May 2022 TZ2 Paper 3 Q1(b)(i), 1 mark

Write down the coordinates of the two points of inflexion on the curve $y^2 = x^3 + 1$, where $x \geq -1$.

The two branches are checked separately: give the upper one first.

```
q4_5a = ...      # the point of inflexion on  
y = +√(x3 + 1)  
q4_5b = ...      # the point of inflexion on  
y = -√(x3 + 1)  
  
verify_bend('4.5(a)', q4_5a, sqrt(x**3 + 1),  
domain=(-0.99, 3), coordinates=True)  
verify_bend('4.5(b)', q4_5b, -sqrt(x**3 + 1),  
domain=(-0.99, 3), coordinates=True)
```

4.6 — May 2022 TZ2 Paper 3 Q1(e), 7 marks

The curve $y^2 = x^3 + x$, $x \geq 0$, has two points of inflexion which, by the symmetry of the curve, share an x -coordinate.

Find the value of this x -coordinate, giving your answer in the form $x = \sqrt{\frac{p\sqrt{3} + q}{r}}$ where $p, q, r \in \mathbb{Z}$.

Exact value. The upper branch $y = \sqrt{x^3 + x}$ carries it.

```
q4_6 = ...      # the x-coordinate of the  
points of inflexion
```

```
verify_bend('4.6', q4_6, sqrt(x**3 + x),  
domain=(0.01, 2))
```

4.7 — May 2023 TZ2 Paper 2 Q12(e), 2 marks

The graph of $y = x\sqrt{\frac{9x^4 - 1}{2}}$ for $\frac{\sqrt{3}}{3} < x < 1$ has a point of inflexion at P .

By sketching the graph of an appropriate derivative of y , determine the x -coordinate of P .

```
q4_7 = ...      # the x-coordinate of the  
inflexion
```

```
verify_bend('4.7', q4_7, x*sqrt((9*x**4 -  
1)/2), domain=(sqrt(3)/3 + 0.001, 1))
```

4.8 — November 2025 TZ1 Paper 1 Q9(d), 2 marks

The function f has $f'(x) = 3x^2 + 12x - 15$, and its second derivative is zero at $x = c = -2$. The sign of f'' on either side of c is $-$ then $+$.

State, with a reason, whether there is a point of inflexion on the graph of $y = f(x)$ at $x = c$.

```
q4_8 = ...          # the kind of point at x = c  
  
verify_nature('4.8', q4_8, integrate(3*x**2 +  
12*x - 15, x), -2, domain=(-9, 5))
```

§ 5. A family with a letter

Two questions, 9 marks. The same work, with a parameter instead of a number, and the words *show that* at the end.

5.1 — November 2023 Paper 3 Q1(e), 5 marks

Consider the family of curves $y = x^3 + ax^2 + b$ for $x \in \mathbb{R}$, $a \neq 0$ and $b \in \mathbb{R}$.

Show that the curve has a point of zero gradient at $P(0, b)$ and a point of zero gradient at $Q\left(-\frac{2a}{3}, \frac{4a^3}{27} + b\right)$.

Enter both points as a list, in terms of a and b .

```
a, b = symbols('a b')

q5_1 = [...]      # the two points of zero
gradient

verify_turning('5.1', q5_1, x**3 + a*x**2 +
b, domain=(-8, 4),
                params={a: (3, 1, 2, 6), b:
(1, 2, 5, 1)})
```

5.2 — May 2021 TZ1 Paper 3 Q1(d), 4 marks

Consider $f(x) = x^3 - 3cx + 2$ for $x \in \mathbb{R}$, where c is a parameter. Given that the graph of $y = f(x)$ has one local maximum point and one local minimum point, show that

(i) the y -coordinate of the local maximum point is $2c^{3/2} + 2$;

(ii) the y -coordinate of the local minimum point is $-2c^{3/2} + 2$.

Enter both points, maximum first, in terms of c .

```
c = symbols('c')

q5_2 = [...]      # the local maximum and the
                  local minimum

verify_turning('5.2', q5_2, x**3 - 3*c*x + 2,
               domain=(-4, 4),
               params={c: (1, 4, Rational(1,
4), 2)}))
```

§ 6. Which side of the axis

Two questions, 2 marks. A different question from the kind of point, and a mark of its own.

6.1 — November 2023 Paper 3 Q1(f) (ii), 1 mark

For the family $y = x^3 + ax^2 + b$ with $a > 0$ and $b > 0$, determine whether each of $P(0, b)$ and $Q\left(-\frac{2a}{3}, \frac{4a^3}{27} + b\right)$ is located above or below the x -axis.

```
q6_1 = [...]      # 'above' or 'below' for the
                    two points, in order

verify_side('6.1', q6_1, x**3 + a*x**2 + b,
            [0, -2*a/3],
            params={a: (3, 1, 2, 6), b: (1,
            2, 5, 1)})
```

6.2 — November 2023 Paper 3 Q1(g) (ii), 1 mark

For the same family with $a < 0$ and $b > 0$, state the conditions on a and b that determine when Q is below the x -axis.

```
q6_2 = ...          # the condition on a and b

def below(a, b):
    place = [spot for spot in stationary(x**3
+ a*x**2 + b, (-20, 20))
            if abs(spot[0]) > 1e-6]
# this is Q, the one away from 0
    return None if not place else place[0][1]
< 0

verify_condition('6.2', q6_2, below,
symbols('a b'), window=(-3, -0.25), steps=10)
```

§ 7. How many

Three questions, 16 marks. Not a question about points but about counting: how many solutions $f'(x) = 0$ has, and how many times the curve meets the axis.

7.1 — May 2021 TZ1 Paper 3 Q1(c), 4 marks

For $f(x) = x^3 - 3cx + 2$, with $f'(x) = 3x^2 - 3c$, find the set of values of c such that the graph of $y = f(x)$ has

- (i)** a point of inflexion with zero gradient;
- (ii)** one local maximum point and one local minimum point;
- (iii)** no points where the gradient is equal to zero.

```

c = symbols('c')

q7_1a = ...      # a point of inflexion with
zero gradient
q7_1b = ...      # one local maximum and one
local minimum
q7_1c = ...      # no points of zero gradient

family = x**3 - 3*c*x + 2

def kinds(value):
    return [word for _, _, word in
stationary(family.subs(c, value), (-5, 5))]

verify_param_set('7.1(i)', q7_1a, lambda v:
kinds(v) == ['inflexion'],
                var=c, window=(-4, 4))
verify_param_set('7.1(ii)', q7_1b, lambda v:
kinds(v) == ['maximum', 'minimum'],
                var=c, window=(-4, 4))
verify_param_set('7.1(iii)', q7_1c, lambda v:
kinds(v) == [], var=c, window=(-4, 4))

```

7.2 — May 2021 TZ1 Paper 3 Q1(e), 6 marks

Hence, for $c > 0$, find the set of values of c such that the graph of $y = f(x) = x^3 - 3cx + 2$ has

- (i)** exactly one x -axis intercept;
- (ii)** exactly two x -axis intercepts;
- (iii)** exactly three x -axis intercepts.

```

q7_2a = ...      # exactly one x-axis
intercept
q7_2b = ...      # exactly two
q7_2c = ...      # exactly three

def meets(value, times):
    if value <= 0:
        return None
    #
    the question speaks only of  $c > 0$ 
    return crossings(x**3 - 3*value*x + 2,
        (-30, 30)) == times

verify_param_set('7.2(i)', q7_2a, lambda v:
meets(v, 1), var=c, window=(0, 4))
verify_param_set('7.2(ii)', q7_2b, lambda v:
meets(v, 2), var=c, window=(0, 4))
verify_param_set('7.2(iii)', q7_2c, lambda v:
meets(v, 3), var=c, window=(0, 4))

```

7.3 — May 2021 TZ1 Paper 3 Q1(f), 6 marks

Consider the function $g(x) = x^3 - 3cx + d$ for $x \in \mathbb{R}$, where $c, d \in \mathbb{R}$.

Find all conditions on c and d such that the graph of $y = g(x)$ has exactly one x -axis intercept, explaining your reasoning.

Write one condition joining the cases:
Or(..., ..., ...) .

```
c, d = symbols('c d')

q7_3 = ...          # all conditions on c and d

def once(c, d):
    return crossings(x**3 - 3*c*x + d, (-30,
30)) == 1

verify_condition('7.3', q7_3, once,
symbols('c d'), window=(-3, 3), steps=12)
```

§ 8. There are none

Two questions, 8 marks. Set the gradient to zero and find that nothing on the curve satisfies it — either because the equation has no real solution, or because its solutions lie outside the domain the question states.

8.1 — November 2022 Paper 1 Q7, 7 marks

Consider the curve with equation $(x^2 + y^2)y^2 = 4x^2$ where $x \geq 0$ and $-2 < y < 2$.

Show that the curve has no local maximum or local minimum points for $x > 0$.

The answer is the list of such points, and it is empty.

```
q8_1 = [...]      # every local maximum or
minimum with x > 0

shape = curve(Eq((x**2 + y**2)*y**2, 4*x**2))

verify_turning('8.1', q8_1, shape, domain=
(0.01, 8))
```

8.2 — May 2022 TZ2 Paper 3 Q1(d)
(ii), 1 mark

For the curve $y^2 = x^3 + x$ with $x \geq 0$, part (d)(i) gives

$$\frac{dy}{dx} = \pm \frac{3x^2 + 1}{2\sqrt{x^3 + x}}, \quad x > 0$$

Hence deduce that the curve $y^2 = x^3 + x$ has no local minimum or maximum points.

```
q8_2 = [...]      # every local maximum or
minimum on the curve

verify_turning('8.2', q8_2, curve(Eq(y**2,
x**3 + x)), domain=(0.01, 4))
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Solutions

§ 1

1.1 $f'(x) = 4x^2 - 16 = 0$ gives $x = \pm 2$; with $p > 0$, $p = 2$ and $q = \frac{32}{3} - 32 = \boxed{-\frac{64}{3}}$.

1.2 $f'(x) = -2 \sin 2x e^{\cos 2x} = 0$ when $\sin 2x = 0$, that is $x = 0, \frac{\pi}{2}, \pi$ inside the interval: $\boxed{(0, e), \left(\frac{\pi}{2}, \frac{1}{e}\right), (\pi, e)}$.

1.3 From the calculator, $A = \boxed{(0.709, 0.640)}$
(0.708519..., 0.639580...).

1.4 Implicitly, $8x + 2y \frac{dy}{dx} - 24 + 4 \frac{dy}{dx} = 0$, so $\frac{dy}{dx} = \frac{4(3-x)}{y+2}$, which is zero at $x = 3$. Then $36 + y^2 - 72 + 4y + 20 = 0$, that is $y^2 + 4y - 16 = 0$ and $y = -2 \pm 2\sqrt{5}$. The maximum takes the upper root: $\boxed{(3, -2 + 2\sqrt{5})} \approx (3, 2.47)$.

1.5 $\boxed{(-1.94, 1.20)}$ is the local maximum and $\boxed{(1.94, -1.20)}$ the local minimum ($\pm 1.94347\dots, \mp 1.20113\dots$).

1.6 $f(x_1) = 12x_1 - x_1^2$ is greatest at $x_1 = \boxed{6}$ by symmetry, and $f(6) = 6 \cdot 6 = \boxed{36}$.

1.7 $s'(t) = \frac{4.3 \cdot 3 \cos \sqrt{3t+5}}{2\sqrt{3t+5}} = 0$ when $\cos \sqrt{3t+5} = 0$. Inside $0 \leq t \leq 10$ the only solution is $\sqrt{3t+5} = \frac{3\pi}{2}$, giving $\boxed{q = 5.74}$ (5.73553...).

1.8 $f'(x) = \frac{1-2x^2}{\sqrt{1-x^2}} = 0$ at $x = \pm \frac{1}{\sqrt{2}}$, where $f = \pm \frac{1}{2}$; at the ends $f(\pm 1) = 0$. So $\boxed{-\frac{1}{2} \leq y \leq \frac{1}{2}}$.

§ 2

2.1 $f''(x) = (4 \sin^2 2x - 4 \cos 2x) e^{\cos 2x}$, so $f''(0) = -4e < 0$, $f''(\frac{\pi}{2}) = \frac{4}{e} > 0$ and $f''(\pi) = -4e < 0$: $\boxed{\text{maximum, minimum, maximum}}$.

2.2 $\frac{d^2y}{dx^2} = \boxed{6x + 2a}$. At $x = 0$ it is $2a > 0$, so P is a $\boxed{\text{minimum}}$; at $x = -\frac{2a}{3}$ it is $-2a < 0$, so Q is a $\boxed{\text{maximum}}$.

2.3 With $a < 0$ the value at $x = 0$ is $2a < 0$, so P is a $\boxed{\text{maximum}}$.

2.4 $V = 4x^3 - 2(a+b)x^2 + abx$, so $\frac{d^2V}{dx^2} = 24x - 4(a+b)$, and at x_m

$$24 \cdot \frac{(a+b) - \sqrt{a^2 - ab + b^2}}{6} - 4(a+b) \\ = -4\sqrt{a^2 - ab + b^2} < 0$$

so x_m gives a local maximum.

§ 3

3.1 f' changes from positive to negative at $x = a$, so the curve rises and then falls: a local maximum.

3.2 $f'_n(-1) = n(-1)^{n-1}(a+2)(a+1)^{n-1}$, and $(a+2)$ and $(a+1)^{n-1}$ are positive.

(i) For even n , $n-1$ is odd and $(-1)^{n-1} = -1$, so $f'_n(-1) < 0$; with $f'_n(0) = 0$ and $f'_n(\frac{a}{4}) > 0$ the gradient runs $-, 0, +$ and $(0, 0)$ is a local minimum.

(ii) For odd n , $n-1$ is even and $f'_n(-1) > 0$; the gradient runs $+, 0, +$ and $(0, 0)$ is a point of inflexion with zero gradient.

§ 4

4.1 The local minimum of f' (or the zero of f'') is at $\boxed{x = -1.60}$ ($-1.59537\dots$).

4.2 $g'(x) = 2(x - r)(x - a) + x^2 - 2ax + a^2 + b^2$, so $g''(x) = 2(x - a) + 2(x - r) + 2x - 2a = 6x - 2r - 4a$, which vanishes at $\boxed{x = \frac{1}{3}(2a + r)}$.

4.3 $\frac{1}{3}(2a + r) = r + \frac{2}{3}(a - r)$, so P is $\boxed{\frac{2}{3}}$ of the horizontal distance from R to A .

4.4 With $a = r$ the formula gives $x = \frac{1}{3}(2r + r) = r$, and $g(r) = (r - r)((r - r)^2 + b^2) = 0$: $\boxed{P(r, 0)}$.

4.5 The curve is $y = \pm\sqrt{x^3 + 1}$, and each branch bends the other way across $x = 0$: $\boxed{(0, 1)}$ and $\boxed{(0, -1)}$.

4.6 Differentiating $2y\frac{dy}{dx} = 3x^2 + 1$ again, $2\left(\frac{dy}{dx}\right)^2 + 2y\frac{d^2y}{dx^2} = 6x$, so at an inflexion $\left(\frac{dy}{dx}\right)^2 = 3x$ and

$$\begin{aligned}
\frac{(3x^2 + 1)^2}{4(x^3 + x)} &= 3x \\
&\implies 3x^4 + 6x^2 - 1 \\
&= 0 \\
&\implies x^2 \\
&= \frac{2\sqrt{3} - 3}{3}
\end{aligned}$$

$$x = \sqrt{\frac{2\sqrt{3} - 3}{3}}$$

$$\approx 0.393 \quad (p = 2, q = -3, r = 3)$$

4.7 The zero of $\frac{d^2y}{dx^2}$ is at $x = 0.656$ (0.655996...).

4.8 f'' changes sign at $x = c$, so the concavity changes: yes, a point of inflexion.
 "Because $f''(c) = 0$ " alone earns R0.

§ 5

5.1 $\frac{dy}{dx} = 3x^2 + 2ax = x(3x + 2a)$, zero at $x = 0$ and $x = -\frac{2a}{3}$. At $x = 0$, $y = b$; at $x = -\frac{2a}{3}$,

$$y = -\frac{8a^3}{27} + \frac{4a^3}{9} + b = \frac{4a^3}{27} + b$$

5.2 $f'(x) = 3x^2 - 3c = 0$ gives $x = \pm\sqrt{c}$, and the maximum is the left one:

$$\begin{aligned} f(-\sqrt{c}) &= -c^{3/2} + 3c^{3/2} + 2 \\ &= \boxed{2c^{3/2} + 2}, & f(\sqrt{c}) \\ &= \boxed{-2c^{3/2} + 2} \end{aligned}$$

§ 6

6.1 $b > 0$ puts P above the axis, and $\frac{4a^3}{27} + b > 0$ for $a > 0$ puts Q above it too: $\boxed{\text{above, above}}$.

6.2 Q is below the axis exactly when its y -coordinate is negative: $\boxed{\frac{4a^3}{27} + b < 0}$.

§ 7

7.1 $3x^2 = 3c$ means $x^2 = c$.

(i) One repeated root, the gradient touching zero: $\boxed{c = 0}$. **(ii)** Two distinct roots: $\boxed{c > 0}$. **(iii)** No real roots: $\boxed{c < 0}$.

7.2 The maximum value $2c^{3/2} + 2$ is always positive, so the minimum value decides.

(i) $-2c^{3/2} + 2 > 0$, that is $\boxed{0 < c < 1}$. **(ii)** $-2c^{3/2} + 2 = 0$, that is $\boxed{c = 1}$. **(iii)** $-2c^{3/2} + 2 < 0$, that is $\boxed{c > 1}$.

7.3 For $c \leq 0$ the derivative $3x^2 - 3c$ is never negative, so g never turns and meets the axis once whatever d is. For $c > 0$ the vertices are $(-\sqrt{c}, 2c^{3/2} + d)$ and $(\sqrt{c}, -2c^{3/2} + d)$, and one intercept means both lie on the same side:

$$\boxed{c \leq 0, \quad \text{or} \quad d > 2c^{3/2}, \quad \text{or} \quad d < -2c^{3/2}}$$

§ 8

8.1 Writing the equation as $x^2y^2 + y^4 = 4x^2$ and differentiating,

$$2xy^2 + 2x^2y \frac{dy}{dx} + 4y^3 \frac{dy}{dx} = 8x$$

At a maximum or minimum $\frac{dy}{dx} = 0$, so $2xy^2 = 8x$ and $x = 0$ or $y = \pm 2$. The domain

says $x > 0$ and $-2 < y < 2$, so nothing survives: there are none.

8.2 $\frac{dy}{dx} = 0$ needs $3x^2 + 1 = 0$, and $3x^2 + 1 \geq 1$ for every real x . So there are none.